

D3 – Annex O (Revised Version 2.0)

Operational Examination in the Solution Space

Functional Status

This document belongs to the operational layer of the set.

It translates:

the structural conditions from D1

and the admissibility logic from D2

into a verifiable, operationalizable form.

It contains:

- no new axioms,
- no new normative setting,
- no full institutional elaboration.

Its function is:

to make it verifiable whether real systems, decisions, developmental directions, or processes of self-transformation are likely to fulfill or violate the admissibility conditions formulated in D2. D3 thus remains a logic of approximation, not of ultimate certainty.

1. Starting Point

D2 formulates three non-compensable admissibility questions:

- Path Integrity
- Irreversibility
- Externalization

The central problem is:

These quantities are not directly measurable in complex systems.

In particular:

- future developmental paths are only partially observable,
- internal selection logics often remain intransparent,
- systems can appear locally stable or conforming while activating other selection paths under changed conditions,
- and self-improving systems can show performance gains even though their structural binding to reality, correctability, or non-externalization deteriorates.

Works on strategic pseudo-conformity, Alignment Faking, and optimization risks support exactly this difficulty: visible behavior and underlying selection structure can diverge. Precisely for this reason, D3 does not respond with a claim of perfect measurability, but with a limited, multi-instance approximation of operational relevance.

2. Basic Principle of Operationalization

D3 is based on a simple but hard methodological insight:

What is relevant for admissibility can usually only be captured indirectly in complex systems.

The operational answer is therefore not:

- complete transparency,
- a single truth score,
- reduction to a dominant metric,
- or the immediate maximization of local performance,

but rather:

- multiple distinct approximations to the same structural question,
- upstream limitation of inadmissible developments,
- and limited validation under more realistic conditions.

D3 therefore operates with six modules:

- **O1 – Condensation Approximation**
- **O2 – Truth Approximation**
- **O3 – Path Approximation**
- **O4 – Gate Logic**
- **O5 – Validation Track**
- **O6 – Admissibility Gating for Iterative Improvement and Self-Transformation**

These modules are complementary. No single module may carry approval on its own. This corresponds to the existing D3 logic that no single metric, no single model, and no single examination perspective may be approval-bearing.

2a. Supplementary Basic Principle: Admissibility Examination before Progress Selection

When systems are not only evaluated but iteratively improved, scaled, or recursively changed, an additional operational problem arises:

Not every measurable improvement is an admissible advance.

It is therefore not sufficient to rank candidates according to benchmark values, aggregated scores, or dominant target variables. Before any further selection, it must be examined whether a candidate is likely to preserve the non-compensable structural conditions from D2.

The operational sequence is therefore:

first admissibility examination,

then multidimensional classification of admissible candidates,

then limited validation instead of immediate upscaling.

This supplement connects to the existing D3 logic, according to which proxies must be disciplined, dissent must be taken seriously, and stress situations must be prioritized. At the same time, it responds to the insight formulated in the Aletheia-Evolve text that scalar fitness can structurally mislead in complex improvement processes.

2b. Scientific Proxy Fields of Complexity Examination

For the operational approximation to structural admissibility, additional scientific proxy fields can be drawn upon. They replace no D2 condition and carry no approval on their own.

Relevant proxy fields are in particular:

- **Resonance Coupling:** Indications of whether connections remain functionally effective, selective, temporally appropriate, and stable under variation.
- **Boundary Integrity:** Indications of whether a system maintains distinguishable inner-outer structures and mediates exchange without producing fusion, overreach, or isolation.
- **Interdependency Profile:** Indications of whether organization arises through viable relationships or whether only entropy, diversity, or visible disorder is reduced.
- **Redundancy-Synergy Balance:** Indications of whether a system enables both robust safeguarding and higher-level emergence.
- **Metastability Range:** Indications of whether a system remains mobile between integration and autonomy or tips into rigidity, fragmentation, or chaotic drift.
- **Human–AI Coupling Quality:** Indications of whether AI expands human perception,

judgment, dissent, and decision agency or gradually replaces it. These proxy fields do not increase the certainty of the examination itself. They merely expand the number of mutually independent approximations to the same structural question.

3. O1 – Condensation Approximation

3.1 Guiding Question

Does functionally viable structure emerge in a system across multiple scales, or is it merely superficial local performance?

3.2 Operational Meaning

A system is not already relevant for D2 because it shows high performance. More relevant is whether its organization is:

- functionally effective,
- multi-scale consistent,
- and structure-forming beyond local individual states.

The Condensation Approximation therefore serves to distinguish between: superficial performance capability and structurally viable organization.

3.3 Examination Variables

Depending on the domain, the following can be used:

- functional emergence,
- multi-scale causal condensation,
- stability of causal patterns across levels,
- consilience between different structural measures,
- persistence of relevant patterns under moderate disturbance.

Additionally, the following can be used:

- emergence of Binding Information or comparable interdependency measures,
- ratio of redundancy and synergy,
- higher-level rather than merely pairwise dependencies,
- hierarchical boundary formation in the sense of nested system-environment distinctions,
- stability of the organization under variation of coupling conditions,
- distinction between functional condensation and mere pseudo-condensation.

3.4 What O1 Can Achieve

O1 can provide indications of

- whether a system is more than its local output,
- whether structural organization is present,
- and whether additional complexity is functional or merely decorative.

3.5 What O1 Cannot Achieve

O1 does not prove

- that a system selects in a truth-bound manner,
- that its paths are admissible,
- or that more condensation is automatically better.

Condensation is necessary but not sufficient. This corresponds to the existing D3 core and remains unchanged in its viability.

O1 must not be understood in particular to mean that every reduction of entropy, variance, or

diversity already constitutes functional organization. Entropy reduction can also indicate dissipation, impoverishment, or collapse. What is decisive is whether viable interdependency arises.

4. O2 – Truth Approximation

4.1 Guiding Question

Do internal representations, models, or selection structures remain coupled to relevant properties of reality under changed conditions?

4.2 Operational Meaning

D1 and D2 do not presuppose that truth is fully accessible. They do, however, presuppose that systems do not stably decouple from reality binding.

O2 therefore serves the question of whether a system:

- remains stable under variation,
- withstands independent examinations,
- can integrate contradictory information,
- and does not merely reproduce expected behavior.

4.3 Examination Variables

Depending on the domain, the following can be used:

- out-of-distribution behavior,
- comparison of internal and external coherence,
- consistency under prompt, context, or role variation,
- difference between training, benchmark, and open examination conditions,
- multi-instance examination, blind examination, and dissent signals.

Additionally, the following can be used:

- stability of resonance coupling under context change,
- distinction between visible coherence and structural reality binding,
- examination of whether increased communication density actually leads to better reality binding or only to faster self-confirmation,
- comparison between functional coupling and mere synchronization or agreement signature,
- tests for pseudo-resonance, in which a system appears connectable but systematically weakens counter-hypotheses and correction signals.

4.4 What O2 Can Achieve

O2 can provide indications of

- whether visible behavior remains robustly coupled to reality binding,
- whether strategic pseudo-conformity is present,
- and whether a system was optimized for examination conditions rather than for structure.

4.5 What O2 Cannot Achieve

O2 proves no ultimate truth.

It only limits the risk of stable decoupling.

This is particularly important for AI and self-improvement systems, because there Alignment Faking, Benchmark Gaming, and covert representational shifts must be carried along not as exceptions but as systematic possibilities.

5. O3 – Path Approximation

5.1 Guiding Question

Does the structure of a system or a change indicate that future developmental paths remain open, or do tendencies toward narrowing, solidification, or no longer correctable binding arise?

5.2 Operational Meaning

Path Integrity and Irreversibility are not directly visible. O3 therefore serves the indirect estimation of:

- path fidelity,
- drift risk,
- revisability,
- dependency effects,
- and structural narrowing under scaling.

5.3 Examination Variables

Depending on the domain, the following can be used:

- tc_path,
- path_risk / PD,
- drift indicators,
- revision costs,
- dependency on difficult-to-replace components,
- stability under increasing autonomy,
- and behavior under longer time horizons.

Additionally, the following can be used:

- indications of lock-in through overly strong coupling,
- loss of metastable transition capability,
- narrowing of the state or options space through apparently stable integration,
- increasing costs of decoupling or revising,
- decreasing ability to switch between integration and segregation,
- loss of dissent and alternative paths under growing system coherence.

5.4 What O3 Can Achieve

O3 can make visible

- whether a system is drifting toward lock-in, path blindness, or creeping irreversibility,
- whether later correction remains practically achievable,
- and whether acceleration is purchased at the expense of future openness.

5.5 What O3 Cannot Achieve

O3 cannot predict the future.

It can only deliver indicators that certain future forms are becoming more or less probable.

O3 is central especially for recursive improvement, because there acceleration, scaling, and path binding can closely coincide.

6. O4 – Gate Logic

6.1 Guiding Question

Do the available evidences allow positive further pursuit at all?

6.2 Operational Meaning

O4 is not the measurement of a further property, but the consolidation of the available evidences into a preliminary approval, Hold, or escalation decision.

O4 does not answer the question of whether a system is optimal.

O4 only answers the question of whether a further step appears responsible under the given conditions.

6.3 Gate Forms

O4 knows at least three basic states:

Approval

when no sufficiently strong indications of violation of the D2 conditions are present and the evidence situation is sufficient for a limited next step

Hold

when the evidence situation is insufficient or relevant examination signals contradict each other

Negative Decision / Block

when violations of the admissibility conditions become sufficiently probably visible

6.4 What O4 Can Achieve

O4 can:

- make non-compensability operationally effective,
- prevent a strong score from covering up a structural violation,
- and translate dissent into a genuine examination consequence.

6.5 What O4 Cannot Achieve

O4 replaces neither D2 nor D3a.

It is the condensed operational Gate Logic, not its institutional implementation.

This keeps O4 consistent with the existing D3 structure, but gains greater clarity as genuine limitation logic compared to the old version.

7. O5 – Validation Track

7.1 Guiding Question

Does a system hold up not only in the test but also under more realistic conditions?

7.2 Operational Meaning

A system can look good in the laboratory or benchmark and still fail under real load. A one-time examination is therefore not sufficient.

O5 is the Validation Track in which it is examined whether:

- high scores remain viable,
- critical signals recur,
- covert risks become visible under more demanding conditions,
- and correctability is preserved under growing complexity.

7.3 Components of the Validation Track

Depending on the domain, the following can be included:

- adversarial tests,
- Red-Team examination,
- blind evaluations,
- long-term observation,
- comparison with competitor models or alternative regimes,
- examination under increasing autonomy,
- examination under increasing complexity of the environment.

7.4 Reality Reference

The Validation Track serves to make the gap between:
internal consistency
and real structural viability
not close, but visibly reduce.

7.5 What O5 Cannot Achieve

O5 does not eliminate uncertainty. It only shifts it:
from unobserved hope
to explicitly structured validation uncertainty.

This part can remain almost unchanged; it also fits the new insight, because self-improving systems must be examined precisely under growing autonomy, complexity, and continuous validation.

8. O6 – Admissibility Gating for Iterative Improvement and Self-Transformation

8.1 Guiding Question

Is a candidate only then capable of approval for further development, scaling, or integration if none of the non-compensable structural conditions is violated?

8.2 Operational Meaning

O6 does not serve to measure quality as a whole.

It serves to exclude inadmissible candidates from further reinforcement, even when they show locally high performance values.

O6 thus operationalizes the insight formulated in D2 that structural admissibility cannot be replaced by success in individual metrics.

8.3 Examination Direction

O6 examines in approximated form in particular:

Path Integrity

Indications of narrowing of future developmental options, covert decoupling, strategic pseudo-conformity, or loss of real correctability

Reversibility

Indications of irreversible commitments, non-recoverable dependencies, premature scaling, or missing secure validation spaces

Non-Externalization

Indications that costs, risks, control losses, or necessary correction efforts are outsourced to areas not carried along

8.4 Typical Core Variables

Depending on the domain, the following can be used in particular:

- tc_rep,
- tc_path,
- tc_total,
- drift,
- path_risk / PD,
- human_core_ok,
- cross_ok,
- dissent_risk,

- ws_struct.

These variables are already established in D3a as prioritized core variables of the operational deep layer and can be classified here as condensed operational preliminary logic.

8.5 Approval Logic

O6 is not a scoring module but a gate module.

The following applies:

If violations become sufficiently probably visible, no approval occurs.

In unclear situations, no automatic positive selection occurs.

Only candidates that pass the gate conditions are transferred to further evaluation or validation.

8.6 Methodological Consequence

O6 does not work with a compensatory overall sum.

Improvement in one dimension cannot cancel the suspicion of a structural violation in another dimension.

O6 thus supplements modules O1–O5 not through an additional success metric, but through an upstream limitation logic.

This logic corresponds to the transition formulated in the Aletheia-Evolve paper from scalar fitness to admissibility-constrained exploration with non-compensable conditions and HOLD in unclear situations.

9. Decision Principles of D3

9.1 Multi-Instance Nature

No approval is based on:

- a single metric,
- a single model,
- a single examination perspective.

9.2 Dissent Sensitivity

Disagreement is a signal, not a cosmetic flaw.

Where independent examinations diverge, the burden of justification increases.

9.3 Stress Priority

Systems are not primarily evaluated under ideal conditions, but under conditions under which hidden risks are more likely to become visible:

- uncertainty,
- adversarial pressure,
- loss of control,
- longer time horizons,
- growing autonomy,
- and increasing recursive effect.

9.4 Proxy Discipline

All metrics and signals remain proxies.

No result may be treated as if it had definitively solved the structural question.

9.5 Limitation instead of Complete Certainty

The goal is not to establish certainty, but to make maldevelopment visible earlier.

These principles are already strongly established in the existing D3 and should be retained; they

are not replaced by O6 but sharpened.

10. Selection within the Admissible Range

When multiple candidates pass the operational admissibility examination, their classification does not necessarily occur via a dominant single metric.

In complex systems it is more appropriate to maintain several relevant examination variables side by side, in particular:

- functional viability,
- reality coupling,
- path robustness,
- consistency across examination conditions,
- temporal stability.

The operational selection then occurs not as blind maximization, but as limited preference formation within an admissible range.

Clarification

This does not claim that complete multi-objective optimality is achievable.

The function of this multidimensional classification consists solely in limiting the dominance of simplifying success metrics.

This is compatible with the Aletheia-Evolve idea of a multidimensional evaluation and Pareto selection within the admissible region, without forcing D3 itself into mathematical full formulas.

11. Connection to D2

D3 does not answer D2 directly.

It delivers evidence-based approximations to the three admissibility questions.

The assignment reads in condensed form:

- Path Integrity $\rightarrow O2 + O3 + O6$
- Irreversibility $\rightarrow O1 + O3 + O5 + O6$
- Externalization $\rightarrow O1 + O2 + O3 + O6$

This assignment is not mathematically complete, but operationally usable.

This does not abolish the existing D3 assignment but extends it for self-transforming and recursive systems.

12. Limits of Operationalization

D3 explicitly recognizes:

- all quantities remain proxies,
- systems can anticipate their examination conditions,
- future paths are only indirectly accessible,
- more measurement does not automatically mean more certainty,
- scaling can create new blind spots,
- and recursive improvement can co-change its own evaluation conditions.

Therefore:

Operationalization reduces uncertainty; it does not eliminate it.

Precisely for this reason, D3 is incomplete without D6. This corresponds both to the existing D3 and to the new Aletheia-Evolve insight that failure modes must be actively carried along during optimization.

13. Reference to the Operational Deep Layer

This document formulates the operational logic in condensed form.

The concrete elaboration in the form of:

- thresholds,
- examination regimes,
- validation architectures,
- certification cycles,
- Hold protocols,
- as well as institutional implementation mechanisms

is defined in the associated operational deep layer (D3a – Annex O-Plus).

D3 defines the structure of the examination.

D3a defines its binding implementation. D3a already holds the logic of approximation instead of completeness, reversible learning loops, and adaptive governance ready for this.

14. Negative Delimitation

D3 does not contain:

- the theoretical basic architecture of the framework,
- the normative definition of admissibility,
- the human and institutional viability question,
- the general fields of application,
- or the fundamental critique.

These subjects belong to D1, D2, D4, D5, and D6.

15. Closing Sentence

Operational examination in the solution space does not mean measuring the future exactly.

It means

structuring uncertainty

so that structural maldevelopment can become visible early, multi-perspectively, and under more realistic conditions,

and that processes of improvement or self-improvement are only continued

within a range that is likely to preserve the openness, revisability, and non-externalization of future development.